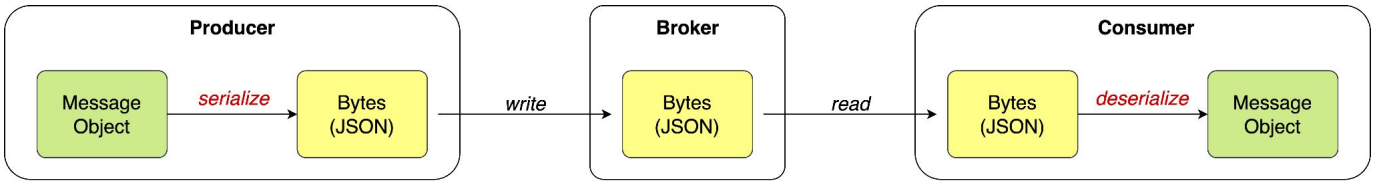
Serialization

“Appear weak when you are strong,
and strong when you are weak.”
– Sun Tzu



1. Introduction

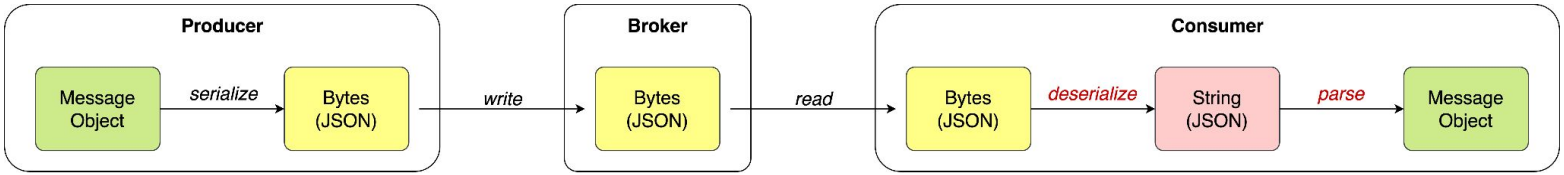
- Serialization is the process of **converting objects or data structures into bytes**.
- Why Serialization?
  - **Preserve the state** of an object so that it **can be recreated** (deserialized) later.
  - **Efficiency for storage and transmission**.
- Deserialization is the reverse process, converting bytes into objects.



2.1. Serialization, Parsing, and Marshaling

	Serialization	**Parsing**	**Marshaling**
Context	**Converting** objects into a storable or transmittable format. A broader concept.	**Analyzing** and converting textual data into a structured format.	A form of serialization often used in RPC
Direction	**Bidirectional**	**Unidirectional** (in theory)	Bidirectional
Purpose	Preserving the state of an object for **storage or transmission**.	Used for **understanding** and processing textual data formats	Prepare objects for RPC.
Use cases	● Caching objects in memory ● Sending data over HTTP or other communication protocols	● Parsing JSON string to an object ● Parsing HTML to a DOM ● Reading configuration files	● Make the remote method call (RPC)

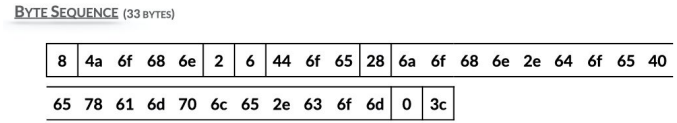
2.2. Serialization vs Parsing



- Mistake: deserializing from bytes to string then parsing from string to an object.
- Best practice: **deserializing from bytes to an object directly.**

3.1. Formats

- Text-based formats:
  - String
  - XML
  - **JSON**
- Binary-based formats:
  - **Avro**
  - Thrift
  - Protobuf
- Others:
  - Int, Float
  - …



Avro Binary Format

3.2. Format Selection

- Factors for selection:
  - **Complexity**
  - **Compatibility**
  - Size
  - Speed
  - …
- Why **JSON**?
  - Simple
  - Compatibility
  - Suite to beginner



4. Configuration

- Serialization is applied to the key and value in records.
- No default values for the serializer, deserializer.
- **Deserializer must be compatible for serializer.**
- 2 ways to configure serializer/deserializer:
  - **Property-based**
    - Pros: Simple
    - Cons: Not type-safety; Complex to customize
  - Instantiating serializers
    - Pros:
      - Customizable for serializers/deserializers
      - Generic type-safety
    - Cons: Manual

